# Lecture 9. Hypothesis Testing (Part 1)

One-Sample Tests and Weighted $K$-Sample Log-Rank Tests

## Contents

**1 Overview** **1**

1.1 One-sample problems . . . 1

1.2 Two- and $K$-sample problems . . . 1

1.3 How to test? . . . 2

**2 One-Sample Test with Right-Censored Data** **2**

2.1 One-sample log-rank test . . . 2

2.2 Harrington–Fleming weights . . . 3

**3 $K$-Sample Tests** **3**

3.1 Hypotheses . . . 3

3.2 Data and notation . . . 3

3.3 Basic idea and the test statistic . . . 4

3.4 Distribution under $H_0$ . . . 4

3.5 Linear dependency and the final statistic . . . 5

3.6 Special case $K = 2$ . . . 5

**4 Choice of the Weight Function** **5**

**5 Example 7.2: Catheter Placement in Kidney Dialysis Patients** **6**

**6 Example 7.4: Three-Group Comparison (Bone Marrow Transplant)** **7**

## 1 Overview

### 1.1 One-sample problems

**Rationale.** A special pattern in a KM curve may suggest a parametric model; parametric models are useful (efficiency; handling more complex sampling schemes), but once adopted we must check whether the model fits the data adequately. Hypotheses:

$$H_0 : S(t) = S_0(t), \qquad H_0 : H(t) = H_0(t), \qquad H_0 : h(t) = h_0(t) \qquad \text{for all } t \le \tau,$$

with $S_0$, $H_0$, $h_0$ *known*. E.g. under $X \sim \text{Exp}(\lambda)$: $H_0(t) = \lambda t$, $h_0(t) = \lambda$, $S_0(t) = e^{-\lambda t}$.

### 1.2 Two- and $K$-sample problems

The study population may be heterogeneous, with groups or key factors relevant to survival. With two groups we compare

$$H_0 : S_1(t) = S_2(t) \iff H_1(t) = H_2(t) \iff h_1(t) = h_2(t), \qquad t \in [0, \tau],$$

equivalent hypotheses with *no* known form. The $K$-sample case is the obvious extension: $H_0 : S_1(t) = \cdots = S_K(t)$, $t \in [0, \tau]$.

### 1.3 How to test?

Without censoring: one-sample $t$ test or Kolmogorov–Smirnov; two-/$K$-sample $t$ tests and ANOVA (approximately normal data) or Wilcoxon rank-sum and Kruskal–Wallis (skewed data). The difficulty is censoring (and truncation); our focus is censoring.

## 2 One-Sample Test with Right-Censored Data

Data $\{(T_i, \delta_i)\}$, $T_i = \min(X_i, C_i)$, $\delta_i = I(X_i \le C_i)$, $X_i$ with hazard $h(t)$. Test $H_0 : h(t) = h_0(t)$, equivalently $H_0 : H(t) = H_0(t)$, for $t \in [0, \tau]$, where $\tau$ is typically the largest observed study time. Notation: $t_1 < \cdots < t_D$ the distinct death times, $d_j = \sum_i I(T_i = t_j, \delta_i = 1)$, $Y_j = Y(t_j) = \sum_i I(T_i \ge t_j)$.

**Definition: One-sample weighted statistic**

Let $W(t)$ be a weight function with $W(t) = 0$ whenever $Y(t) = 0$. Compare the Nelson–Aalen increments with the hypothesised hazard:

$$Z(\tau) = O(\tau) - E(\tau) = \sum_{j=1}^{D} W(t_j) \frac{d_j}{Y(t_j)} - \int_0^\tau W(s) h_0(s)\, \mathrm{d}s.$$

Under $H_0$, its variance is

$$V[Z(\tau)] = \int_0^\tau W^2(s) \frac{h_0(s)}{Y(s)}\, \mathrm{d}s,$$

and for large samples $Z(\tau)^2 / V[Z(\tau)] \sim \chi^2_1$.

**Remark: Intuition**

$\tilde{H}(t) = \sum_{t_j \le t} d_j / Y_j$ estimates $H(t)$; the statistic is a weighted distance $\int W \, \mathrm{d}(\tilde{H} - H_0)$. Each increment $d_j / Y_j$ has (Poisson) variance $\approx h_0 \, \mathrm{d}s / Y$, which gives $V[Z]$.

- The $\chi^2$ statistic only serves two-sided alternatives.
- One-sided: for $H_A : h(t) > h_0(t)$ for some $t \le \tau$, reject for large positive $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$; for $H_A : h(t) < h_0(t)$, reject for large negative $T_1$.

### 2.1 One-sample log-rank test

**Theorem: $W(t) = Y(t)$: observed vs. expected**

Allow left truncation: $T_i$ = time on study, $L_i$ = delayed entry time. With $\tau$ the largest time on study,

$$O(\tau) = \sum_{j=1}^{D} Y(t_j) \frac{d_j}{Y(t_j)} = \sum_{j=1}^{D} \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1) = \sum_{i=1}^{n} I(\delta_i = 1),$$

$$E(\tau) = \int_0^\tau Y(s) h_0(s)\, \mathrm{d}s = \sum_{i=1}^{n} \int_0^\tau I(L_i \le s \le T_i) h_0(s)\, \mathrm{d}s = \sum_{i=1}^{n} [H_0(T_i) - H_0(L_i)].$$

$O$ = observed number of events at or before $\tau$; $E$ = expected number under $H_0$. Also $V[Z(\tau)] = \int_0^\tau Y(s)h_0(s)\,\mathrm{d}s = E(\tau)$, so the test statistic is $(O-E)^2/E$.

### 2.2 Harrington–Fleming weights

$$W_{HF}(t) = Y(t)S_0(t)^p[1-S_0(t)]^q, \qquad p,q \geq 0, \quad S_0(t) = \exp[-H_0(t)].$$

- $p \gg q$: more weight on **early** departures from $H_0$;
- $q \gg p$: more weight on **late** departures;
- $p = q$: more weight on departures in the **mid-range**;
- $p = q = 0$: the one-sample log-rank test.

**Example: IBCSG Trial 6**

International Breast Cancer Study Group: a multi-centre trial of adjuvant chemotherapy after surgical removal of the tumour; 735 subjects. The NA cumulative hazard plotted against time is roughly a straight line up to 100 months, suggesting an exponential distribution; eyeballing the slope gives $\hat{\lambda} \approx 0.005$. Formal one-sample log-rank tests ($W = Y$):

| $H_0$ | $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$ | $p$-value |
|---|---|---|
| $H_0(t) = 0.0045t$ | $-0.002$ | 0.998 |
| $H_0(t) = 0.004t$ | 1.8 | 0.07 |
| $H_0(t) = 0.005t$ | $-1.4$ | 0.16 |
| $H_0(t) = 0.006t$ | $-4.1$ | $< 0.01$ |

$H_0(t) = 0.0045t$ fits best. (Sign check: a larger hypothesised rate inflates $E$, making $O - E$ negative.)

## 3 $K$-Sample Tests

### 3.1 Hypotheses

$H_0 : h_1(t) = h_2(t) = \cdots = h_K(t) \;\; \forall t \leq \tau$ vs. $H_A :$ at least one $h_k(t)$ differs for some $t \leq \tau$.

Equivalently in terms of $H_k$ or $S_k$. Here $\tau$ is the largest time at which all groups have at least one subject at risk, typically $\tau = \min_k t_{\max,k}$. The alternative is *global*: reject if at least one population differs from the others at some time.

### 3.2 Data and notation

Right-censored data $\{(T_{ik}, \delta_{ik}),\ k = 1,\ldots,K,\ i = 1,\ldots,n_k\}$. Pool all data and let $0 < t_1 < \cdots < t_D$ be the distinct death times in the pooled sample:

- $d_j$, $d_{jk}$: deaths at $t_j$ in the pooled sample / in group $k$;
- $Y_j$, $Y_{jk}$: number at risk at $t_j$ in the pooled sample / in group $k$.

### 3.3 Basic idea and the test statistic

Under $H_0$ the hazards are equal but unknown, so both the pooled and within-group NA estimators estimate the same $H$:

$$\tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j} \to \int_0^t h(u)\,\mathrm{d}u, \qquad \tilde{H}_k(t) = \sum_{t_j \le t} I(Y_{jk} > 0)\frac{d_{jk}}{Y_{jk}} \to \int_0^t h(u)\,\mathrm{d}u.$$

Build a test from a weighted difference between $\tilde{H}_k$ and $\tilde{H}$.

**Definition: Weighted $K$-sample statistic**

With $W_k(t) \ge 0$ a random weight, zero when $Y_j = 0$ or $Y_{jk} = 0$:

$$Z_k(\tau) = \sum_{j=1}^{D} W_k(t_j)\Big\{\frac{d_{jk}}{Y_{jk}} - \frac{d_j}{Y_j}\Big\}.$$

Typically $W_k(t_j) = Y_{jk}W(t_j)$ with $W(\cdot)$ common to all groups, giving

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\} \quad = \sum_j W(t_j)\,(\text{observed} - \text{expected deaths in group } k).$$

### 3.4 Distribution under $H_0$

**Theorem: Asymptotic normality (via martingale theory)**

Under $H_0$, $(Z_1(\tau), \ldots, Z_K(\tau)) \approx \mathrm{MVN}(\mathbf{0}, \Sigma)$, $\Sigma = (\sigma_{kk'})_{K\times K}$, with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k = 1, \ldots, K,$$

$$\hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k \ne k'.$$

**Derivation: hypergeometric origin of the variance**

At $t_j$, condition on the margins: $Y_j$ at risk split into $(Y_{j1}, \ldots, Y_{jK})$ with $d_j$ deaths. Under $H_0$ the deaths are a random draw of $d_j$ out of $Y_j$ without replacement, so $(d_{j1}, \ldots, d_{jK})$ is multivariate hypergeometric with

$$\mathrm{E}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}, \quad \mathrm{Var}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}, \quad \mathrm{Cov}(d_{jk}, d_{jk'}) = -d_j\frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}.$$

The factor $(Y_j - d_j)/(Y_j - 1)$ is the finite-population correction (it equals 1 when $d_j = 1$). Summing over (approximately uncorrelated) death times gives $\hat{\sigma}_{kk'}$.

### 3.5 Linear dependency and the final statistic

**Theorem:** $\sum_k Z_k(\tau) = 0$

$$\sum_{k=1}^{K} Z_k(\tau) = \sum_{j=1}^{D} W(t_j) \sum_{k=1}^{K} \Big\{ d_{jk} - d_j \frac{Y_{jk}}{Y_j} \Big\} = \sum_{j=1}^{D} W(t_j)(d_j - d_j) = 0,$$

since $\sum_k d_{jk} = d_j$ and $\sum_k Y_{jk} = Y_j$. Hence $Z_1, \ldots, Z_K$ are linearly dependent and $\Sigma$ is singular.

Eliminate the redundancy by dropping one (arbitrary) $Z_K(\tau)$:

$$(Z_1, \ldots, Z_{K-1}) \approx \text{MVN}(\mathbf{0}, \Sigma_{(K-1)\times(K-1)}),$$

and

$$\chi^2 = (Z_1(\tau), \ldots, Z_{K-1}(\tau))\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, (Z_1(\tau), \ldots, Z_{K-1}(\tau))^T \overset{H_0}{\sim} \chi^2_{K-1}.$$

Reject at level $\alpha$ if $\chi^2 > \chi^2(K-1, 1-\alpha)$. (The value does not depend on which $Z_k$ is dropped.)

### 3.6 Special case $K = 2$

$$Z = \frac{Z_1(\tau)}{\sqrt{\hat{V}(Z_1(\tau))}} = \frac{\sum_{j=1}^{D} W(t_j)(d_{j1} - Y_{j1} d_j / Y_j)}{\sqrt{\sum_{j=1}^{D} W^2(t_j) \frac{Y_{j1}}{Y_j}(1 - \frac{Y_{j1}}{Y_j}) \frac{Y_j - d_j}{Y_j - 1} d_j}} \overset{H_0}{\approx} N(0,1).$$

For $K = 2$, a two-sided level-$\alpha$ test rejects if $|Z| > Z_{1-\alpha/2}$; one-sided tests are possible. For $K > 2$ we must use the $\chi^2$ test, which only handles two-sided alternatives.

## 4 Choice of the Weight Function

Different weights change the power against specific alternatives; they are extremely well studied. All have the form $W_k(t_j) = Y_{jk} W(t_j)$.

1. **Log-rank**: $W(t) = 1$. Optimal power against **proportional hazards** alternatives $h_1(t) = c_2 h_2(t) = \cdots = c_K h_K(t)$.
2. **Gehan** (generalised Wilcoxon): $W(t) = Y(t) = \sum_k \sum_i I(T_{ik} \geq t)$, so $Z_k(\tau) = \sum_j (Y_j d_{jk} - d_j Y_{jk})$. Commonly used.
3. **Tarone–Ware**: $W(t) = f(Y(t))$ with known increasing $f$; $f(u) = u$ is Gehan, the original Tarone–Ware uses $f(u) = u^{1/2}$, also $f(u) = \log u$. Gives relatively larger weight to death times with larger risk sets.
4. **Peto–Peto**: $W(t) = \tilde{S}(t)$, $\tilde{S}(t) = \prod_{t_j \leq t} (1 - \frac{d_j}{Y_j + 1})$, a KM-like estimator from the pooled sample.
5. **Modified Peto–Peto** (Andersen, 1982): $W(t) = \tilde{S}(t) Y(t)/(Y(t) + 1)$. Both Peto–Peto tests depend on the pooled survival function.
6. **Fleming–Harrington**: $W(t_j) = \hat{S}(t_{j-1})^p [1 - \hat{S}(t_{j-1})]^q$, $p, q \geq 0$, $\hat{S}$ the pooled KM.
   - $p = q = 0$: log-rank;
   - $p > 0, q = 0$: larger weights early $\Rightarrow$ sensitive to early differences among hazards;
   - $p = 0, q > 0$: larger weights late $\Rightarrow$ sensitive to late differences.

**Remark: Why the left-continuous $\hat{S}(t_{j-1})$**

The weight at $t_j$ must be *predictable* — known just before $t_j$ — so it uses survival information up to $t_{j-1}$, not the deaths at $t_j$ themselves. This is what makes $Z_k$ a martingale-based statistic.

## 5 Example 7.2: Catheter Placement in Kidney Dialysis Patients

Clinical trial (K&M Section 1.4) comparing time to cutaneous exit-site infection for catheters placed surgically (group 1) vs. percutaneously (group 2). From the KM curves (Fig. 7.1), the surgical group does better early, but the percutaneous group has higher infection-free probability later (about 0.8 vs. 0.2 near month 27). The curves *cross*, so the choice of weight matters:

- weighting early times heavily (e.g. Gehan) may suggest the surgical group is better;
- weighting late times heavily may suggest the percutaneous group is better.

Relative weights (Fig. 7.2): Gehan and Tarone–Ware decrease steeply with time; log-rank is flat; Fleming–Harrington with $q > 0$ increase with time.

**Example: Log-rank computation (K&M Table 7.2, selected rows)**

| $t_j$ | $Y_{j1}$ | $d_{j1}$ | $Y_{j2}$ | $d_{j2}$ | $Y_j$ | $d_j$ | $Y_{j1}\frac{d_j}{Y_j}$ | $d_{j1} - Y_{j1}\frac{d_j}{Y_j}$ | $\frac{Y_{j1}}{Y_j}\left(1-\frac{Y_{j1}}{Y_j}\right)\frac{Y_j-d_j}{Y_j-1}d_j$ |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 | 43 | 0 | 76 | 6 | 119 | 6 | 2.168 | −2.168 | 1.326 |
| 1.5 | 43 | 1 | 60 | 0 | 103 | 1 | 0.417 | 0.583 | 0.243 |
| 2.5 | 42 | 0 | 56 | 2 | 98 | 2 | 0.857 | −0.857 | 0.485 |
| 4.5 | 36 | 2 | 43 | 0 | 79 | 2 | 0.911 | 1.089 | 0.490 |
| | | | | | | | ⋮ | | |
| 26.5 | 2 | 1 | 3 | 0 | 5 | 1 | 0.400 | 0.600 | 0.240 |
| SUM | | 15 | | 11 | | 26 | 11.036 | 3.964 | 6.211 |

So $Z_1(\tau) = 15 - 11.036 = 3.964$, $\hat{\sigma}_{11} = 6.211$, $\chi^2 = 3.964^2/6.211 = 2.53$, $p = 0.112$: no significant difference by the log-rank test.

**Example: Comparison of two-sample tests (K&M Table 7.3)**

| Test | $W(t_j)$ | $Z_1(\tau)$ | $\sigma^2_{11}$ | $\chi^2$ | $p$-value |
|---|---|---|---|---|---|
| Log-rank | 1 | 3.96 | 6.21 | 2.53 | 0.112 |
| Gehan | $Y_j$ | −9 | 38862 | 0.002 | 0.964 |
| Tarone–Ware | $Y_j^{1/2}$ | 13.20 | 432.83 | 0.40 | 0.526 |
| Peto–Peto | $\tilde{S}(t_j)$ | 2.47 | 4.36 | 1.40 | 0.237 |
| Modified Peto–Peto | $\tilde{S}(t_j)Y_j/(Y_j+1)$ | 2.31 | 4.20 | 1.28 | 0.259 |
| FH $p=0, q=1$ | $1-\hat{S}(t_{j-1})$ | 1.41 | 0.21 | 9.67 | 0.002 |
| FH $p=1, q=0$ | $\hat{S}(t_{j-1})$ | 2.55 | 4.69 | 1.39 | 0.239 |
| FH $p=1, q=1$ | $\hat{S}(t_{j-1})[1-\hat{S}(t_{j-1})]$ | 1.02 | 0.11 | 9.83 | 0.002 |
| FH $p=0.5, q=0.5$ | $\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{0.5}$ | 2.47 | 0.66 | 9.28 | 0.002 |
| FH $p=0.5, q=2$ | $\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^2$ | 0.32 | 0.01 | 8.18 | 0.004 |

**Remark: Comments**

- Gehan's weight puts very heavy weight on early comparisons, leading to a *negative* $Z_1(\tau)$.
- Fleming–Harrington tests with $q > 0$ weight late comparisons and are significant, because the two curves diverge for large $t$.
- In practice, the choice of weight is an important decision and should be made *before* looking at the data.
- The log-rank test ($W = 1$) is the most popular, followed by Gehan (generalised Wilcoxon rank-sum); both are in most statistical packages.

## 6 Example 7.4: Three-Group Comparison (Bone Marrow Transplant)

Disease-free survival of leukaemia patients after bone marrow transplantation: Group 1 = 38 ALL, Group 2 = 54 AML low-risk, Group 3 = 45 AML high-risk. Test $H_0 : S_1(t) = S_2(t) = S_3(t)$, $t \le \tau = 2204$ days.

**Example: Log-rank test with $K = 3$**

Using the first two statistics, $Z_1(2204) = 2.148$, $Z_2(2204) = -14.966$, with

$$\hat{\Sigma} = \begin{pmatrix} 15.9552 & -10.3451 \\ -10.3451 & 20.3398 \end{pmatrix}, \qquad \chi^2 = (2.148,\ -14.966)\,\hat{\Sigma}^{-1} \begin{pmatrix} 2.148 \\ -14.966 \end{pmatrix} = 13.8037.$$

On $K - 1 = 2$ df, $p \approx 0.001$.

**Derivation: evaluating the quadratic form**

$\det \hat{\Sigma} = 15.9552(20.3398) - 10.3451^2 = 324.53 - 107.02 = 217.51$, and $\hat{\Sigma}^{-1} = \frac{1}{217.51}\begin{pmatrix} 20.3398 & 10.3451 \\ 10.3451 & 15.9552 \end{pmatrix}$. Then

$$\begin{aligned} \chi^2 &= \frac{1}{217.51}\Big[20.3398(2.148)^2 + 2(10.3451)(2.148)(-14.966) + 15.9552(-14.966)^2\Big] \\ &= \frac{93.85 - 665.14 + 3573.73}{217.51} \approx 13.80. \end{aligned}$$

Other weights: Gehan $\chi^2 = 16.24$ ($p = 0.0003$); Tarone–Ware 15.65 ($p = 0.0004$); FH $(1,0)$ 15.67 ($p = 0.0004$); FH $(0,1)$ 6.11 ($p = 0.047$); FH $(1,1)$ 9.93 ($p = 0.007$). The results are robust: all tests suggest that the disease-free survival curves differ among the three groups.